Global facet selection for accelerated velocity constraints

Abstract

We study accelerated optimization with velocity constraints on an affine polyhedron, using the original selected-constraint method of Muehlebach and Jordan. Suppose that the objective is globally smooth and strongly convex, and that exactly one defining inequality is active at the minimizer, with a positive multiplier. From every feasible zero-velocity start, we give an explicit index after which that optimal inequality is selected at every step. The proof retains the strict quadratic remainder in the existing energy argument and uses a signed energy to exclude all late returns to the feasible side of the optimal facet. It gives global boundedness and an accelerated Lagrangian-gap bound with the actual incoming velocity. An explicit later index guarantees that all other inequalities are inactive and the optimal velocity inequality binds permanently. A quadratic refinement and the failure of Lyapunov stability describe the scope of these conclusions.

Keywords: accelerated optimization; velocity constraints; polyhedral constraints; finite selection

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1 Introduction

Velocity constraints provide a way to impose inequalities through a projected direction while allowing the position iterates to be infeasible. The accelerated method of Muehlebach and Jordan [2] selects the inequalities that are active or violated at the current position. Their constant-step accelerated rate is proved for a modification using all inequalities; analogous asymptotic rates for the original selected-constraint method are conjectured. For affine constraints, the curvature correction in that modification vanishes, but the difference between the two selection rules remains.

This note proves a global selection bound for the original accelerated recurrence when the optimizer has exactly one active defining inequality and its multiplier is positive. Every other inequality has strict slack there. The starting position may be any feasible point, with zero initial velocity. In particular, the polyhedron may be compact, with arbitrarily narrow or oblique additional faces. The central estimate rules out every sufficiently late return to the strictly feasible side of the optimal facet. It therefore supplies the permanent support inclusion needed by the existing accelerated energy argument.

The energy and the quadratic coefficients used below come from the proof of Theorem 4 in [2], specifically the calculation following its Eq. (27). Conditional on inclusion of the optimal multiplier support, that proof already gives the one-step accelerated contraction. Our contribution is the explicit global bound that makes this condition hold along the original selected-set trajectory. We retain its strict remainder and use the signed objective

energy before support inclusion. The global convergence and eventual selection mechanism also has a first-order antecedent in Muehlebach and Jordan [1, Proposition 3 and Section 5, Claim 6]. At the endpoint where the momentum is zero, the compact-polytope qualitative conclusion is already covered by that first-order theory.

Permanent selection is not finite arrival on the feasible facet. Once the velocity inequality binds, a negative normal coordinate can approach zero geometrically while remaining negative at every finite step. The Lagrangian gap, rather than a necessarily nonnegative objective gap at infeasible iterates, is the quantity controlled here. The results do not include extra zero-multiplier inequalities active at the optimizer, multiple active optimal inequalities, or nonlinear constraints.

1.1 The recurrence and the main theorem

Let $d \geq 1$, $0 < m \leq 1$, and let $f : \mathbb{R}^d \to \mathbb{R}$ be continuously differentiable, globally m-strongly convex, with globally 1-Lipschitz gradient. Suppose

$$\nabla f(0) = \ell e_1, \qquad \ell > 0, \qquad C = \{x : g_i(x) \geq 0,\ 1 \leq i \leq p\}, \tag{1}$$

where

$$g_1(x) = x_1, \qquad g_i(x) = a_i^T x + c_i, \quad c_i > 0,\ a_i \neq 0 \quad (2 \leq i \leq p). \tag{2}$$

Write $a_1 = e_1$ and $c_1 = 0$. Constant redundant inequalities are omitted. The origin is the unique minimizer of f over C, since strong convexity gives

$$f(x) - f(0) \geq \ell x_1 + \frac{m}{2} \|x\|^2 \qquad (x \in C).$$

After translation, an orthogonal change of coordinates, and positive rescaling of the active inequality, these assumptions describe an affine polyhedron with exactly one active defining inequality at the optimizer and a positive multiplier.

Set

$$s = \sqrt{m}, \qquad q = \frac{1}{1+s}, \qquad \alpha = 1 - q = sq, \qquad \beta = 2q - 1 = \frac{1-s}{1+s}. \tag{3}$$

For $x_0 \in C$ and $u_0 = 0$, the iteration is

$$\begin{aligned} I_k &= \{i : g_i(x_k) \leq 0\}, & y_k &= x_k + \beta u_k, \\ u_{k+1} &= \operatorname*{argmin}_{v \in \mathbb{R}^d} \frac{1}{2} \|v - \beta u_k + \nabla f(y_k)\|^2 & &\text{subject to } a_i^T v + \alpha g_i(x_k) \geq 0 \quad (i \in I_k), \\ x_{k+1} &= x_k + u_{k+1}. \end{aligned} \tag{4}$$

In particular, equality is included in the selection rule. This is the affine, zero-restitution specialization of the original Eq. (8) in [2].

Define the optimal Lagrangian and two energies by

$$\begin{aligned} l(x) &= f(x) - \ell x_1, \\ V(x,u) &= \frac{1}{2}\left\|\alpha x + q u\right\|^2 + l(x) - l(0), \\ E(x,u) &= V(x,u) + \ell x_1 = \frac{1}{2}\left\|\alpha x + q u\right\|^2 + f(x) - f(0). \end{aligned} \tag{5}$$

We use $V_k = V(x_k, u_k)$ and $E_k = E(x_k, u_k)$. The energy V is positive definite, whereas E can be negative. Introduce

$$S = \begin{pmatrix} \alpha m - \alpha^3 & \alpha(\beta m + \alpha^2) \\ \alpha(\beta m + \alpha^2) & \alpha - \alpha^2 - \alpha^3 + \beta^2 m \end{pmatrix}, \qquad \eta = \frac{\det S}{2 \operatorname{tr} S}, \tag{6}$$

and

$$\begin{aligned} \rho &= \min_{2 \le i \le p} \frac{c_i}{\|a_i\|}, \qquad F = 2\sqrt{1+\beta^2}, \\ D &= \eta \min\{\rho^2, \ell^2/F^2\}, \qquad C_m = 2\big[m^{-1} + (2+\sqrt{m})^2\big], \\ V_{\mathrm{core}} &= \min\{m\rho^2/8, \ell^2/(64 C_m)\}. \end{aligned} \tag{7}$$

When $p = 1$, set $\rho = \infty$ and interpret $m\rho^2/8$ as infinity. We will prove $S \succ 0$, so $\eta, D, V_{\mathrm{core}}$ are strictly positive and finite.

Theorem 1. *Under (1)–(4), every velocity subproblem has a unique solution. For every $k \ge 0$,*

$$E_k \le q^k E_0, \qquad V_k \le 2q^k E_0 + \frac{2\ell^2}{m}. \tag{8}$$

If $E_0 > 0$, define

$$\begin{aligned} K &= \max\left\{1, \left\lceil \frac{\log(E_0/D)}{\log(1/q)} \right\rceil \right\}, \qquad W = 2q^K E_0 + 2\ell^2/m, \\ N &= \max\left\{0, \left\lceil \frac{\log(W/V_{\mathrm{core}})}{\log(1/q)} \right\rceil \right\}, \quad J = K + N. \end{aligned} \tag{9}$$

Then the following assertions hold.

1. *At every step, $x_{k+1,1} > 0$ implies $qE_k > D$. Consequently,*

$$x_{k,1} \le 0 \qquad \text{for every } k \ge K. \tag{10}$$

2. *The actual state converges to $(0,0)$, and, for every $k \ge K$,*

$$\frac{m}{2}\left\|x_k\right\|^2 \le l(x_k) - l(0) \le V_k \le q^{k-K} V_K \le q^{k-K} W. \tag{11}$$

   Moreover, $\|x_k\|^2 + \|u_k\|^2 \le C_m V_k$ at every index.

3. *For every $k \ge J$, one has $I_k = \{1\}$. The multiplier of the selected velocity inequality is at least $\ell/2$, and*

$$u_{k+1,1} = -\alpha x_{k,1}, \qquad x_{k+1,1} = q x_{k,1}. \tag{12}$$

If $E_0 = 0$, then $x_0 = 0$ and the state remains zero. In that case take $K = J = 0$, omit W, N, and the selection and binding assertions still hold.

Equivalently, K may be any integer at least one with $q^K E_0 \le D$, and N any non-negative integer with $q^N W \le V_{\mathrm{core}}$. Thus equality at either threshold is permitted. The constants involve the optimal multiplier and the inactive-inequality slack scale. They are mathematical entry bounds, not an asserted stopping test using only observed iterates, and they deteriorate as those quantities tend to zero. No angle or width bound is imposed on the polyhedron.

For an objective f_{orig} with smoothness L and strong convexity μ, the normalization is

$$\begin{gathered} f = f_{\mathrm{orig}}/L, \qquad m = \mu/L, \qquad u_k = u_{k,\mathrm{orig}}/\sqrt{L}, \qquad T = 1/\sqrt{L}, \\ \alpha_{\mathrm{orig}} = \delta_{\mathrm{orig}} = \sqrt{L}\,\alpha, \qquad \beta_{\mathrm{orig}} = \beta/\sqrt{L}. \end{gathered} \tag{13}$$

Indeed, the original projection target is $(1 - 2\delta_{\mathrm{orig}}T)u_{k,\mathrm{orig}} - T\nabla f_{\mathrm{orig}}(x_k + \beta_{\mathrm{orig}}u_{k,\mathrm{orig}})$. Dividing the projected velocity and its affine constraints by $\sqrt{L}$ gives (4), while $x_{k+1} = x_k + Tu_{k+1,\mathrm{orig}}$. This retains the original velocity scaling and selection rule.

We prove the exact energy identity in Section 2, and use it to exclude all late returns and establish Theorem 1 in Section 3. Section 4 gives the stability limitation and the elementary quadratic refinement.

2 The strict energy identity

Since $\nabla l(0) = 0$ and l is m-strongly convex,

$$l(x) - l(0) \ge \frac{m}{2}\|x\|^2, \qquad V(x,u) \ge 0, \tag{14}$$

with equality in $V = 0$ only at $(x,u) = (0,0)$. In particular,

$$\|x\| \le \sqrt{2V/m}, \qquad \|u\| \le q^{-1}\sqrt{2V} + (\alpha/q)\sqrt{2V/m} = (2+s)\sqrt{2V}.$$

Squaring and adding proves

$$\|x\|^2 + \|u\|^2 \le C_m V(x,u). \tag{15}$$

Fix any state (x,u), with no feasibility assumption. The selected velocity set is nonempty: $v_* = -\alpha x$ satisfies every selected inequality because

$$a_i^T v_* + \alpha g_i(x) = \alpha c_i \ge 0.$$

It is a closed convex polyhedron, so the Euclidean projection exists and is unique. Write v for that projection, $z = x + v$, and $y = x + \beta u$. Equivalently, z is the projection of $y - \nabla f(y)$ onto

$$D(x) = \{z : g_i(z) \ge q g_i(x),\ i \in I(x)\}.$$

The point qx belongs to $D(x)$, since $g_i(qx) = qg_i(x) + \alpha c_i$. Thus, with

$$r = z - y + \nabla f(y), \tag{16}$$

the projection variational inequality yields $r^T(qx - z) \geq 0$.

We can also record the exact multiplier form. The selected constraints are strictly feasible: v_* already makes all selected indices $i \geq 2$ strict, and, if index 1 is selected, $v_* + \varepsilon e_1$ makes it strict while preserving the other strict inequalities for sufficiently small $\varepsilon > 0$. Hence there are multipliers $\lambda_i \geq 0$ with

$$r = \sum_{i \in I(x)} \lambda_i a_i, \qquad \lambda_i \big(a_i^T v + \alpha g_i(x)\big) = 0.$$

The identity $a_i^T(qx - z) = \alpha c_i - (a_i^T v + \alpha g_i(x))$ therefore gives

$$r^T(qx - z) = \alpha \sum_{i \in I(x)} \lambda_i c_i \geq 0. \tag{17}$$

This calculation includes all selected extra faces, without assumptions on their angles or previous activity. If $x_1 \leq 0$, selection of index 1 also gives

$$z_1 \geq q x_1. \tag{18}$$

Define the Bregman difference and the quadratic remainder by

$$B_f(a, b) = f(a) - f(b) - \nabla f(b)^T(a - b), \qquad R_m(x, u) = \frac{1}{2} \begin{pmatrix} x \\ u \end{pmatrix}^T (S \otimes I_d) \begin{pmatrix} x \\ u \end{pmatrix}.$$

The remainder coefficients in (6) are the ones retained in the proof of [2, Theorem 4]. The following identity makes their role in the signed energy explicit.

Lemma 1. *For every step of* (4)*,*

$$\begin{aligned} qE(x, u) - E(z, v) = R_m(x, u) + r^T(qx - z) \\ + \big[\tfrac{1}{2} \|z - y\|^2 - B_f(z, y)\big] \\ + \alpha\big[B_f(0, y) - \tfrac{m}{2} \|y\|^2\big] \\ + q\big[B_f(x, y) - \tfrac{m}{2}\beta^2 \|u\|^2\big]. \end{aligned} \tag{19}$$

Every term after R_m is nonnegative, and

$$S \succ 0, \qquad R_m(x, u) \geq \eta(\|x\|^2 + \|u\|^2). \tag{20}$$

Consequently, $E_{k+1} + R_m(x_k, u_k) \leq qE_k$ at every step.

Proof. We first verify the algebra, including the signs of the signed energy. Put $g = \nabla f(y)$. Since $\alpha + q = 1$, expansion of the Bregman terms gives

$$\begin{aligned} - B_f(z, y) + \alpha B_f(0, y) + q B_f(x, y) \\ = q(f(x) - f(0)) - (f(z) - f(0)) + g^T(z - qx). \end{aligned} \tag{21}$$

Adding $r^T(qx - z)$ cancels the gradient term and leaves $-(z-y)^T(z-qx)$. Moreover,

$$\frac{1}{2}\|z-y\|^2 - (z-y)^T(z-qx) = \frac{1}{2}\|y-qx\|^2 - \frac{1}{2}\|z-qx\|^2. \tag{22}$$

Here $y - qx = \alpha x + \beta u$, $x - y = -\beta u$, and $\alpha z + qv = z - qx$. Therefore (21)–(22) reduce (19) to the purely quadratic identity

$$\begin{aligned} 2R_m(x,u) = q\|\alpha x + qu\|^2 - \|\alpha x + \beta u\|^2 \\ + m\alpha\|x + \beta u\|^2 + mq\beta^2\|u\|^2. \end{aligned} \tag{23}$$

Expanding its right side gives the coefficients of $\|x\|^2$, $2x^Tu$, and $\|u\|^2$, respectively:

$$\begin{aligned} q\alpha^2 - \alpha^2 + m\alpha &= \alpha m - \alpha^3, \\ \alpha(q^2 - \beta + m\beta) &= \alpha(\alpha^2 + m\beta), \\ q^3 - \beta^2 + m(\alpha+q)\beta^2 &= \alpha - \alpha^2 - \alpha^3 + m\beta^2. \end{aligned}$$

We used $q^2 - \beta = \alpha^2$ and $q^3 - \beta^2 = \alpha - \alpha^2 - \alpha^3$. These are exactly the entries of (6), proving the identity.

Smoothness and strong convexity give, respectively,

$$B_f(z,y) \le \tfrac{1}{2}\|z-y\|^2, \qquad B_f(a,b) \ge \tfrac{m}{2}\|a-b\|^2.$$

Since $x - y = -\beta u$, all three bracketed terms are nonnegative. The projection term is nonnegative by (17).

For strict positivity, substitute $m = s^2$, $\alpha = s/(1+s)$, and $\beta = (1-s)/(1+s)$. Direct expansion gives

$$\begin{aligned} S_{11} &= \frac{s^4(s+2)}{(1+s)^3}, & S_{12} &= \frac{s^3(2-s^2)}{(1+s)^3}, \\ S_{22} &= \frac{s(s^4 - s^3 - 2s^2 + 2s + 1)}{(1+s)^3}. \end{aligned}$$

Multiplying the diagonal entries and subtracting S_{12}^2 yields

$$\det S = \frac{s^5(s^3 - s^2 - s + 2)}{(1+s)^5} = \frac{s^5[1 + (s-1)^2(s+1)]}{(1+s)^5} > 0. \tag{24}$$

Thus $S \succ 0$ by Sylvester's criterion. Its least eigenvalue satisfies

$$\lambda_{\min}(S) = \frac{\det S}{\lambda_{\max}(S)} \ge \frac{\det S}{\operatorname{tr} S}.$$

The same bound applied coordinatewise proves (20). Substitution into (19) completes the proof. ∎

At $m = 1$ these formulas give

$$q = \alpha = \tfrac{1}{2}, \qquad \beta = 0, \qquad S = \begin{pmatrix} 3/8 & 1/8 \\ 1/8 & 1/8 \end{pmatrix}, \qquad \eta = 1/32.$$

Thus the strict remainder does not degenerate at the zero-momentum endpoint.

Iterating Lemma 1 gives $E_k \le q^k E_0$ without requiring $E_k \ge 0$. To bound the positive energy, use (14) and Young's inequality:

$$\ell|x_1| \le \ell\sqrt{2V/m} \le \frac{V}{2} + \frac{\ell^2}{m}.$$

It follows that $E \ge V/2 - \ell^2/m$, proving (8). In particular, the whole state is bounded by (15), even before any conclusion about permanent selection.

3 Global return and permanent selection

The strict remainder turns a positive next normal coordinate into a lower bound on the current signed energy. This is the step that forces inclusion of the optimal inequality from a global start.

Lemma 2. *At any step of* (4)*,*

$$z_1 > 0 \quad \Longrightarrow \quad qE(x,u) > D. \tag{25}$$

Proof. Put $h = \min\{\rho, \ell/F\}$, so $D = \eta h^2$. If $z_1 > 0$, then $E(z,v) = V(z,v) + \ell z_1 > 0$. Lemma 1 consequently gives

$$qE(x,u) > R_m(x,u) \ge \eta(\|x\|^2 + \|u\|^2). \tag{26}$$

If $\|x\| \ge \rho$, the last quantity is at least D, which proves the assertion in that case.

If $\|x\| < \rho$, every extra inequality is strictly positive, since

$$g_i(x) \ge c_i - \|a_i\|\,\|x\| > 0 \qquad (i \ge 2).$$

The actual next normal coordinate is therefore

$$z_1 = \begin{cases} \max\{qx_1, y_1 - \nabla_1 f(y)\}, & x_1 \le 0, \\ y_1 - \nabla_1 f(y), & x_1 > 0. \end{cases} \tag{27}$$

Because $qx_1 \le 0$ in the first case, $z_1 > 0$ implies $y_1 - \nabla_1 f(y) > 0$ in either case. On the other hand,

$$\begin{aligned} y_1 - \nabla_1 f(y) &\le |y_1| + |\nabla_1 f(y) - \ell| - \ell \\ &\le 2\,\|y\| - \ell \\ &\le F\sqrt{\|x\|^2 + \|u\|^2} - \ell. \end{aligned}$$

Here we used $\nabla_1 f(0) = \ell$, gradient Lipschitzness, and the bound

$$\|x + \beta u\| \le \sqrt{1+\beta^2}\sqrt{\|x\|^2 + \|u\|^2}.$$

It follows that the state norm is strictly greater than $\ell/F \geq h$. Equation (26) again yields $qE(x,u) > D$. The equality case $\|x\| = \rho$ was included in the first case, so no extra face was discarded at that boundary. ∎

Proof of Theorem 1. Existence, uniqueness, the global bounds, and the state estimate have already been established. Feasibility of x_0 and $u_0 = 0$ imply

$$E_0 = \tfrac{1}{2}\alpha^2 \|x_0\|^2 + l(x_0) - l(0) + \ell x_{0,1} \geq 0.$$

Equality forces $x_0 = 0$. At the zero state only index 1 is selected; the projection of $-\ell e_1$ onto $v_1 \geq 0$ is zero. Thus this state is fixed, its multiplier is ℓ, and the assertions with $K = J = 0$ follow.

Assume $E_0 > 0$ and choose K as in (9). For every $k \geq K$,

$$qE_{k-1} \leq q^k E_0 \leq q^K E_0 \leq D.$$

Applying Lemma 2 to step $k-1$ proves $x_{k,1} \leq 0$ for every such k. This argument rules out all future positive normal positions, regardless of any earlier crossings or activity of the other faces.

At any step with $x_1 \leq 0$, combine (18) with Lemma 1 to obtain

$$\begin{aligned} qV(x,u) - V(z,v) &= qE(x,u) - E(z,v) + \ell(z_1 - qx_1) \\ &\geq R_m(x,u). \end{aligned} \tag{28}$$

This is the conditional one-step contraction underlying [2, Theorem 4]; it holds with any additional selected faces and with the current velocity. Since $x_{k,1} \leq 0$ for every $k \geq K$, it can now be iterated along the actual trajectory. Equations (14), (8), and (28) give (11), including the bound $V_K \leq W$. Equation (15) then proves $(x_k, u_k) \to (0,0)$. In particular, u_K has not been set to zero or otherwise changed.

For all $k \geq J = K + N$, the same bound gives

$$V_k \leq q^{k-K} W \leq q^N W \leq V_{\text{core}}. \tag{29}$$

If $p > 1$, the first part of the core threshold and (14) imply $\|x_k\| \leq \rho/2$. Hence

$$g_i(x_k) \geq c_i - \|a_i\| \, \rho/2 \geq c_i/2 > 0 \qquad (i \geq 2,\ k \geq J). \tag{30}$$

Together with (10), this proves $I_k = \{1\}$ for every $k \geq J$. This conclusion is immediate when $p = 1$.

For this one-constraint projection the multiplier is

$$\lambda_k = \max\{0, \Delta_k\}, \qquad \Delta_k = \nabla_1 f(x_k + \beta u_k) - \beta u_{k,1} - \alpha x_{k,1}.$$

The second core threshold and (15) give $\sqrt{\|x_k\|^2 + \|u_k\|^2} \leq \ell/8$. Moreover,

$$\begin{aligned} \Delta_k &\geq \ell - \|x_k + \beta u_k\| - \beta \|u_k\| - \alpha \|x_k\| \\ &\geq \ell - (1+\alpha)\|x_k\| - 2\beta \|u_k\| \\ &\geq \ell - 4\sqrt{\|x_k\|^2 + \|u_k\|^2} \geq \ell/2. \end{aligned}$$

The loose constant 4 is valid because $0 < \alpha \leq 1/2$ and $0 \leq \beta < 1$. Thus $\lambda_k \geq \ell/2 > 0$ at every index $k \geq J$. Complementarity forces $u_{k+1,1} + \alpha x_{k,1} = 0$, proving (12). ∎

The proof requires no compactness assumption on C. Arbitrarily small strict slacks only enlarge the entry indices. The role of a single active optimal inequality is more substantial: a positive next normal position makes $E(z, v) = V(z, v) + \ell z_1$ positive. Omission of one inequality among several active optimal inequalities would not give the corresponding sign for their multiplier-weighted sum.

4 Consequences and scope

The theorem proves global attraction from feasible zero-velocity starts, but this attraction does not imply Lyapunov stability, even relative to that class of starts.

Proposition 4.1. The fixed state $(0, 0)$ of (4) is not Lyapunov stable relative to initial states $(x_0, 0)$ with $x_0 \in C$.

Proof. For all sufficiently small $t > 0$, $x_0 = te_1$ makes every defining inequality strictly positive. Thus I_0 is empty, and

$$u_1 = -\nabla f(te_1), \qquad x_1 = te_1 - \nabla f(te_1).$$

Continuity of the gradient gives

$$(x_1, u_1) \longrightarrow (-\ell e_1, -\ell e_1) \qquad (t \downarrow 0),$$

whereas the exact zero start stays fixed. Feasible zero-velocity starts of arbitrarily small norm therefore leave the state-space ball of radius $\ell/2$ at their first step. ∎

For quadratics, permanent binding reduces the tail to a forced linear recurrence. The resulting refinement is a consequence of global selection, rather than a separate mechanism for proving it.

Proposition 4.2. In addition to the assumptions of Theorem 1, suppose

$$f(x) = \tfrac{1}{2} x^T H x + \ell x_1, \qquad H = H^T, \qquad mI \preceq H \preceq I.$$

Then

$$\|(x_{J+n}, u_{J+n})\| = O(q^n), \qquad l(x_{J+n}) - l(0) = O(q^{2n}) \qquad (n \to \infty). \tag{31}$$

If $x_{J,1} < 0$, then

$$\lim_{k \to \infty} [l(x_k) - l(0)]^{1/k} = q^2. \tag{32}$$

If $x_{J,1} = 0$, the normal position stays zero, and no matching q^2 lower rate is asserted.

Proof. Use the actual state at J and write

$$x_{J+n} = (a_n, t_n), \qquad u_{J+n} = (b_n, w_n), \qquad H = \begin{pmatrix} h_{11} & h^T \\ h & B \end{pmatrix}.$$

The tangential coordinates are absent when $d = 1$, in which case the normal formulas below already prove the result. From (12),

$$a_n = q^n a_0 \quad (n \geq 0), \qquad b_n = -\alpha q^{n-1} a_0 = -s a_n \quad (n \geq 1). \tag{33}$$

The incoming velocity b_0 need not satisfy the second relation. Its effect on the first tangential step is retained in the actual values (t_1, w_1).

For $n \geq 1$, set $\gamma = 1 - \beta s$, so $a_n + \beta b_n = \gamma q^n a_0$. The projection changes only the normal component. The tangential dynamics are consequently

$$\begin{pmatrix} t_{n+1} \\ w_{n+1} \end{pmatrix} = T \begin{pmatrix} t_n \\ w_n \end{pmatrix} - \gamma a_0 q^n \begin{pmatrix} h \\ h \end{pmatrix}, \qquad T = \begin{pmatrix} I - B & \beta(I - B) \\ -B & \beta(I - B) \end{pmatrix}. \tag{34}$$

The principal block obeys $mI \preceq B \preceq I$. Orthogonal diagonalization of B decomposes T into two-dimensional blocks, indexed by an eigenvalue $\nu \in [m, 1]$, with characteristic polynomial

$$p_\nu(r) = r^2 - 2q(1 - \nu)r + \beta(1 - \nu). \tag{35}$$

Its discriminant is $4q^2(1 - \nu)(m - \nu) \leq 0$.

First suppose $0 < m < 1$ and put $r_* = 1 - s > 0$. At $\nu = m$ the repeated root is r_*. For $m < \nu < 1$, the roots are conjugate with modulus

$$\sqrt{\beta(1 - \nu)} \leq \sqrt{\beta(1 - m)} = r_*.$$

At $\nu = 1$ the block is nilpotent of order two. The Jordan formula for each repeated-root block, and diagonalization of each distinct-root block over $\mathbb{C}$, therefore give a finite constant C_H such that

$$\left\| T^j \right\| \leq C_H (j + 1) r_*^j \qquad (j \geq 0). \tag{36}$$

There are only finitely many blocks, so a common constant exists; the finitely many nonzero powers of a nilpotent block are absorbed because $r_* > 0$. This explicitly includes the repeated root at $\nu = m$.

Crucially, $r_*/q = 1 - m < 1$. Iterating (34) from $n = 1$ gives, for $n \geq 2$,

$$\begin{pmatrix} t_n \\ w_n \end{pmatrix} = T^{n-1} \begin{pmatrix} t_1 \\ w_1 \end{pmatrix} - \gamma a_0 q^{n-1} \sum_{j=0}^{n-2} q^{-j} T^j \begin{pmatrix} h \\ h \end{pmatrix}.$$

The homogeneous term is $O(q^n)$, since $n(r_*/q)^{n-1}$ is bounded. The norm of the forcing sum, including its prefactor, is bounded by a constant times

$$q^{n-1} \sum_{j=0}^{n-2} (j + 1)(r_*/q)^j \leq \frac{q^{n-1}}{(1 - r_*/q)^2} = O(q^n). \tag{37}$$

Thus $(t_n, w_n) = O(q^n)$. In particular, the repeated-root polynomial factor does not change the rate. Nonresonance can also be checked directly from

$$p_\nu(q) = q^2[(1+m)\nu - m] \geq q^2 m^2 > 0.$$

If $m = 1$, then $H = I$, $\beta = 0$, and $h = 0$. The tangent update is $t_{n+1} = 0$, $w_{n+1} = -t_n$, so the entire tangential state vanishes for $n \geq 2$. This treats the zero repeated-root endpoint without applying (36) with $r_* = 0$.

Together with (33), these estimates prove the state bound in (31). Since $l(x) - l(0) = x^T H x/2 \leq \|x\|^2/2$, they prove its gap bound as well. If $a_0 = x_{J,1} < 0$, strong convexity supplies the matching lower bound

$$l(x_{J+n}) - l(0) \geq \frac{m}{2} a_0^2 q^{2n} > 0.$$

Taking kth roots, with $k = J + n$, proves (32). If $a_0 = 0$, the normal position stays zero by (33); after the first tangential step there is no forcing. For $0 < m < 1$, the remaining homogeneous state is $O((n+1)(1-\sqrt{m})^n)$, and at $m = 1$ it vanishes after the finite transient above. This explains the absence of a matching q^2 assertion in that case. ∎

References

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